24F_CST8116 Assignment 2

Software Development Process, Object Oriented program with decisions and loops.

Instructions

The Software Development Process as presented by Cay Horstmann [1] will be used as the basis for this lab assignment.

1) Understand the problem
2) Develop and Describe an Algorithm
3) Test Algorithm with Simple Inputs
4) Translate the Algorithm into Java
5) Compile and Test Your Program

Overview

- A coin cell battery maker needs to keep track of the quality of the manufactured batteries.
- Each produced battery is measured (diameter and height) and then the dimensions need to be verified using a computer program.
- You are asked to write a computer program where the user can enter the dimensions and the program will determine if both of the dimensions are acceptable, based on the tolerances below:

Dimension	Declared Size [mm]	Tolerances [mm]
Diameter	3.5	±0.1
Height	10.3	±0.3

- The program will keep on prompting for additional battery dimensions until the user indicates they are done. The program will then report how many batteries were out of tolerance.
- Users can make mistakes, so the program should not crash if text is entered instead of numbers when expected. This would be communicated as an error to the user.
- A junior programmer started the project but could not complete it, a senior programmer reviewed the project and then provided some //ToDo comments and has now passed the project to you to complete it.

IMPORTANT:

- You are not permitted to change the structure of the provided classes or methods
- You are not permitted to change the class **User** and the class **Battery**. Only **Driver** and **BatteryChecker**.
- **Your solution must be Object-Oriented – if it is not, then the maximum mark for this assignment is 15%**
- **If you do not submit all 4 Java Source files (regardless if modified), then the maximum mark for this assignment is 10%.**

Task 1: Understand the problem

- Carefully examine the starter code, some of the program's requirements are already implemented for you!
- Examine the starter code, read the ToDo comments, briefly outline in general how you will code the 2 methods below. Provide enough information so that it's clear to others what functionality each method performs:

(1) `checkWithinTolerance()` of class `BatteryChecker`

(2) `main()` of class `Driver`

Task 2: Develop and Describe an Algorithm (UML Class Diagram, flowcharts)

- Document the four classes you have been provided using detailed UML Class diagrams.
  - Remember, static class members should be underlined within a UML Class Diagram.
  - Do not forget to document the different versions of EPSILON when you add them.
  - Carefully observe the access modifiers in class User.
  - **You are not permitted to change the structure of the provided classes or methods**, if in doubt contact your lab professor.

- Create flow-charts only for the 2 methods listed in Task 1.
  - Your main method must use an instance of class **Battery**, an instance of class **BatteryChecker** and use the static class **User**, as described in its documentation.
  - You can use any decision structure (except the conditional operator, see note below) or loop for this program, however, consider that we don't know in advance how many times the loop will run.
  - All user input must be accomplished using the static class User (Do not modify User.java in any manner).

  - **The Conditional Operator (question mark) is not accepted in place of a decision structure for marks in this assignment. Do not use it.**

    - **Example of Conditional Operator: result = (number < 10) ? "less than" : (number > 10) ? "greater than" : "equal to";**

- Place the detailed UML Class Diagrams and flow charts in your MS Word document.

- See lecture materials weeks 9 and 10, as well as the required readings for guidance.

Task 3: Test Algorithm with Simple Inputs

Table 1: Test Plan for method main for the repetition structure logic (looping tests)

- Each row should represent a separate run of the program
- Observe the first two rows of input testing and ensure that you can trace through your flowcharts and obtain the same results
- Complete the table using the indicated inputs in rows 3, 4 and 5 in the submission template

Input	Expected Output	Actual Output	Description
3.5 10.3 yes	Enter battery dimensions (expected: D: 3.50 , H: 10.30) Enter measured diameter: Enter measured height: Battery: diameter 3.50, height 10.30 The battery is within tolerance. Program by Thomas Anderson Continue checking? (e.g.: YES) yes Enter battery dimensions (expected: D: 3.50 , H: 10.30)	Enter battery dimensions (expected: D: 3.50 , H: 10.30) Enter measured diameter: Enter measured height: Battery: diameter 3.50, height 10.30 The battery is within tolerance. Program by Thomas Anderson Continue checking? (e.g.: YES) yes Enter battery dimensions (expected: D: 3.50 , H: 10.30)	A hand trace of the flowchart for method main shows that the logic of the program has the correct sequence of steps, and that the program continues as the user entered yes
3.5 10.3 no	Enter battery dimensions (expected: D: 3.50 , H: 10.30) Enter measured diameter: Enter measured height: Battery: diameter 3.50, height 10.30 The battery is within tolerance. Program by Thomas Anderson Continue checking? (e.g.: YES) Total out of tolerance: 0	Enter battery dimensions (expected: D: 3.50 , H: 10.30) Enter measured diameter: Enter measured height: Battery: diameter 3.50, height 10.30 The battery is within tolerance. Program by Thomas Anderson Continue checking? (e.g.: YES) Total out of tolerance: 0	A hand trace of the flowchart for method main shows that the logic of the program has the correct sequence of steps, and that the program ends when the user entered no
3.5 10.3 YES			
3.5 10.3 NO			
3.5 10.3 maybe			

Table 2: Test Plan for method main for verifying the battery tolerance check.

- Use the table below as a start, but check more cases, the sample table checks length only
- In general, these are the required tests (Boundary Tests)
  - **Tolerance tests for diameter**: below tolerance, at lower tolerance, within tolerance, at upper tolerance, above tolerance
  - **Tolerance tests for height**: below tolerance, at lower tolerance, within tolerance, at upper tolerance, above tolerance
  - **Negative entries tests** for each of the dimensions
- You are provided sample **Tolerance Tests** for **diameter**. Complete the table with in your submission template with:
  - **2 tolerance tests** for **height** (choose 2 from the tolerance tests listed above, e.g. “within tolerance”)
  - **1 negative entry test** for **height**

- Highlight in your testing table the relevant items you are testing and validating, as shown in the samples:

Input	Expected Output	Actual Output	Description
3 10.3	Enter battery dimensions (expected: D: 3.50 , H: 10.30) Enter measured diameter: Enter measured height: Battery: diameter 3.00, height 10.30 The battery is not within tolerance. Program by Thomas Anderson Continue checking? (e.g.: YES)	Enter battery dimensions (expected: D: 3.50 , H: 10.30) Enter measured diameter: Enter measured height: Battery: diameter 3.00, height 10.30 The battery is not within tolerance. Program by Thomas Anderson Continue checking? (e.g.: YES)	Using a hand trace of the flowchart of the checkWithinTolerance method of class BatteryChecker for diameter below tolerance is functioning correctly as the difference is 0.5
3.39 10.3	Enter battery dimensions (expected: D: 3.50 , H: 10.30) Enter measured diameter: Enter measured height: Battery: diameter 3.39, height 10.30 The battery is not within tolerance. Program by Thomas Anderson Continue checking? (e.g.: YES)	Enter battery dimensions (expected: D: 3.50 , H: 10.30) Enter measured diameter: Enter measured height: Battery: diameter 3.39, height 10.30 The battery is not within tolerance. Program by Thomas Anderson Continue checking? (e.g.: YES)	Using a hand trace of the flowchart of the checkWithinTolerance method of class BatteryChecker for diameter at lower tolerance is functioning correctly as the difference is 0.11
3.51 10.3	Enter battery dimensions (expected: D: 3.50 , H: 10.30) Enter measured diameter: Enter measured height: Battery: diameter 3.51, height 10.30 The battery is within tolerance. Program by Thomas Anderson Continue checking? (e.g.: YES)	Enter battery dimensions (expected: D: 3.50 , H: 10.30) Enter measured diameter: Enter measured height: Battery: diameter 3.51, height 10.30 The battery is within tolerance. Program by Thomas Anderson Continue checking? (e.g.: YES)	Using a hand trace of the flowchart of the checkWithinTolerance method of class BatteryChecker for diameter within tolerance is functioning correctly as the difference is 0.01
3.9 10.3	Enter battery dimensions (expected: D: 3.50 , H: 10.30) Enter measured diameter: Enter measured height: Battery: diameter 3.90, height 10.30 The battery is not within tolerance. Program by Thomas Anderson Continue checking? (e.g.: YES)	Enter battery dimensions (expected: D: 3.50 , H: 10.30) Enter measured diameter: Enter measured height: Battery: diameter 3.90, height 10.30 The battery is not within tolerance. Program by Thomas Anderson Continue checking? (e.g.: YES)	Using a hand trace of the flowchart of the checkWithinTolerance method of class BatteryChecker for diameter at upper tolerance is functioning correctly as the difference is 0.4
3.61 10.3	Enter battery dimensions (expected: D: 3.50 , H: 10.30) Enter measured diameter: Enter measured height: Battery: diameter 3.61, height 10.30 The battery is not within tolerance. Program by Thomas Anderson Continue checking? (e.g.: YES)	Enter battery dimensions (expected: D: 3.50 , H: 10.30) Enter measured diameter: Enter measured height: Battery: diameter 3.61, height 10.30 The battery is not within tolerance. Program by Thomas Anderson Continue checking? (e.g.: YES)	Using a hand trace of the flowchart of the checkWithinTolerance method of class BatteryChecker for diameter above tolerance is functioning correctly as the difference is 0.11

Input	Expected Output	Actual Output	Description
-3.5 10.3	Enter battery dimensions (expected: D: 3.50 , H: 10.30) Enter measured diameter: Enter measured height: Battery: diameter -3.50, height 10.30 The battery is not within tolerance. Program by Thomas Anderson Continue checking? (e.g.: YES)	Enter battery dimensions (expected: D: 3.50 , H: 10.30) Enter measured diameter: Enter measured height: Battery: diameter -3.50, height 10.30 The battery is not within tolerance. Program by Thomas Anderson Continue checking? (e.g.: YES)	Using a hand trace of the flowchart of the checkWithinTolerance method of class BatteryChecker for negative diameter is functioning correctly as the difference is 7

Task 4: Translate the Algorithm into Java

- Follow the ToDo sections in the starter code. Do not add additional functionality to the starter code – e.g. there is no requirement to loop if the user enters an invalid entry when asked "Continue Program? (yes / no)"
- Do not change any of the code in class User or class Battery. Make sure you submit all java files regardless.
- Use your flowchart to assist you in completing the methods you have been asked to complete.
- Type brief comments in the source code files for the top of the file, and each class header, each constructor and method header where indicated.
- Reminder, your program must be object-oriented, must use the starter code, and all your source code files must be submitted.
- You may remove the instructional comments as you implement/complete your source code files, replace with your own comment sections where directed.

Task 5: Compile and Test Your Program - Screenshots

- Run your program and take at least **5** screen shots of the running program that validate 5 tests of the Algorithm Test Table (the one provided in Task 3's Table 2 and continued in your submission).
- You only need to include Eclipse's console window in your screenshot. Make sure your name is displayed as part of the program output.

Submission Requirements

- Upload your MS Word document as well as all Java file(s) to the Brightspace submission area by the due date. (See Brightspace for due date).
- Do not include a screen shot of the .java files unless your Lab Professor has that as a submission standard.
- Follow your lab professor's instructions regarding lab submissions for their lab section.
- Before submitting, carefully read the "Grading" section in the last page to avoid losing marks for incomplete submissions. Missing Java files or not using OOP will result in *very low marks*!

Microsoft Word Document Format

Use the template word document

Starter Code

The following .java files are provided.

- Files:
  - `Driver.java`
  - `Battery.java`
  - `BatteryChecker.java`
  - `User.java`

Appendix: Sample of Program Run (your program output may use different wording but have very similar flow)

```
<terminated> Driver (3) [Java Application] /opt/eclipse/plugins/org.e
Enter battery dimensions (expected: D: 3.50 , H: 10.30)
Enter measured diameter: 3.51
Enter measured height: 10.29
Battery: diameter 3.51, height 10.29
The battery is within tolerance.
Program by Thomas Anderson
Continue checking? (e.g.: YES) Yes
Enter battery dimensions (expected: D: 3.50 , H: 10.30)
Enter measured diameter: 3
Enter measured height: 10.3
Battery: diameter 3.00, height 10.30
The battery is not within tolerance.
Program by Thomas Anderson
Continue checking? (e.g.: YES) yes
Enter battery dimensions (expected: D: 3.50 , H: 10.30)
Enter measured diameter: three
Invalid input. Enter a number: 3
Enter measured height: 10.3
Battery: diameter 3.00, height 10.30
The battery is not within tolerance.
Program by Thomas Anderson
Continue checking? (e.g.: YES) But I'm hungry
Total out of tolerance: 2
```

Note that the default settings in Eclipse use a different colour for user input.

Grading

IMPORTANT:

- You are not permitted to change the structure of the provided classes or methods
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- **Your solution must be Object-Oriented – if it is not, then the maximum mark for this assignment is 15%**
- **If you do not submit all 4 Java Source files (regardless if modified), then the maximum mark for this assignment is 10%.**

Criteria	Missing / Poor (0)	Below Expectations (1)	Meets Expectations (2)
Algorithm: UML Class Diagrams	Missing or poorly done or missing code files.	Class diagram(s) are not in correct format, properties and methods may not be assigned correctly to the classes and/or the diagrams do not follow the starter code provided.	Class diagram(s) are correct format, properties and methods are assigned to appropriate classes, based on the word problem and provided starter code.
Algorithm: Flowchart	Missing or poorly done or missing code files.	Not in correct format and/or steps are not in an order that produces correct results. Use of repetition structure(s) and/or selection structure(s) partly correct.	Correct format, steps are in order that produces correct results. Use of repetition structure(s) and/or selection structure(s) is correct.
Test Plan: Algorithm	Missing or poorly done or missing code files.	Does not have correct table format as seen in lecture notes and lab exercises, and/or does not test program using suggested tests.	Has correct table format as seen in lecture and lab exercises, tests program using suggested tests.
Test Plan: Program	Missing or poorly done or unchanged copy of algorithm test table, or missing code files.	Follows from algorithm test table but does not have updated descriptions for all of the tests.	Follows from algorithm test table, does have updated descriptions for all of the tests.
Source Code: *.java file(s) Comments and Conventions	Missing or poorly done or missing code files or is starter code with no modifications.	File comment header with student name is present. Class and/or class-member (constructors, methods) are missing comment headers. Loosely follows Java coding conventions for identifiers, indentation.	File comment header with student full name is present. Class and/or class-member (constructors, methods) have comment headers. Closely follows Java coding conventions for identifiers, indentation.
Source Code: *.java file(s) Program Structure and Logic	Missing or poorly done or missing code files or is starter code with no modifications.	Program may have small syntax mistakes or produces incorrect output. Use of if, if-else, or switch, and/or while or do-while not fully correct (has syntax, runtime, and or logic errors).	Program has correct syntax and program logic that produces correct output. Use of if, if-else, or switch, and/or while or do-while is correct (no syntax, runtime, or logic errors).
5 Screen Shots of Program Execution in Eclipse Console Window	Missing or poorly done or missing student name in screen shot or missing code files.	Screen shot may not show the program running to completion. Student name is present but may not be part of program output.	Screen shot shows program running to completion and shows student name as part of program output.

References

[1] Cay Horstmann. (2019). Big Java Early Objects. 7th Ed. Wiley.